

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Industrial and Systems Engineering

2. Subject Name: Production Planning and Control **L-T-P:** 3-1-0 **Credits:** 4

3. Pre-requisites: IM21201 - Operations Research - I

4. Syllabus and reference books:

Syllabus:

Demand forecasting: Long and Short term demand forecasting methods, Regression analysis and smoothing methods; Estimation of trend, cycle, and seasonality components, Holt's model and Winter's model; Introduction to Box-Jenkins forecasting methods; Analysis of forecast error. Production-distribution system design: Introduction to facility design, Facility layout and types of layout problems; Product and equipment analysis; Traditional approaches to facility layout; Algorithms for layout problems; Multiple plant production facility design. Aggregate planning and master production scheduling: Aggregate planning and Aggregation techniques, Aggregate capacity scheduling; Master production scheduling; Bill of materials; Rough-Cut capacity planning; Material requirements planning. Operations scheduling and control: Basic sequencing and scheduling techniques, Techniques for scheduling line-flow and batch processes; Sequencing using dispatching rules, Sequencing jobs using Johnson's rule; Dispatching in MRP; Progress chasing and Updating of production schedules. Design of production planning and control systems: System design for continuous and intermittent production systems; Integration of master production, Material requirement and Shop scheduling systems; Industry 4.0.

Reference Books:

- 1) Rein Peterson, and Edward A. Silver, Decision Systems for Inventory Management and Production Planning, John Wiley and Sons.
- 2) Stephen N. Chapman, The Fundamentals of Production Planning and Control, Pearson Education, 2006.
- 3) Sunderesh S. Heragu, Facilities Design, CRC Press, Taylor & Francis, 2016.
- 4) Seetharama L. Narasimhan, Dennis W. McLeavey, and Peter J. Billington, Production Planning and Inventory Control, 2nd Edition, Prentice-Hall of India.
- 5) Charles C. Holt, Franco Modigliani, John F. Muth, and Herbert A. Simon, Planning Production, Inventories, and Workforce, Prentice-Hall.
- 6) Alasdair Gilchrist, Industry 4.0: The Industrial Internet of Things, Apress, Thailand.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Demand forecasting: Long and Short term demand forecasting methods, Regression analysis and smoothing methods; Estimation of trend, cycle, and seasonality components, Holt's model and Winter's model; Introduction to Box-Jenkins forecasting methods; Analysis of forecast error.	8
2.	Production-distribution system design: Introduction to facility design, Facility layout and types of layout problems; Product and equipment analysis; Traditional approaches to facility layout; Algorithms for layout problems; Multiple plant production facility design.	8
3.	Aggregate planning and master production scheduling: Aggregate planning and Aggregation techniques, Aggregate capacity scheduling; Master production scheduling; Bill of materials; Rough-Cut capacity planning; Material requirements planning.	6
4.	Operations scheduling and control: Basic sequencing and scheduling techniques, Techniques for scheduling line-flow and batch processes; Sequencing using dispatching rules, Sequencing jobs using Johnson's rule; Dispatching in MRP; Progress chasing and Updating of production schedules.	7
5.	Design of production planning and control systems: System design for continuous and intermittent production systems; Integration of master production, Material requirement and Shop scheduling systems; Industry 4.0.	5
Total number of hours		34